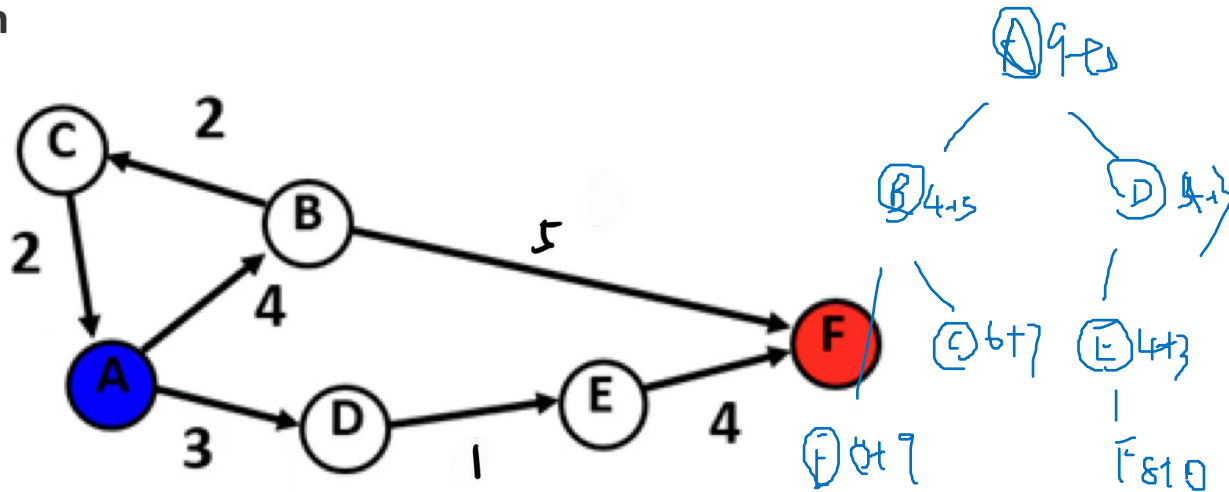
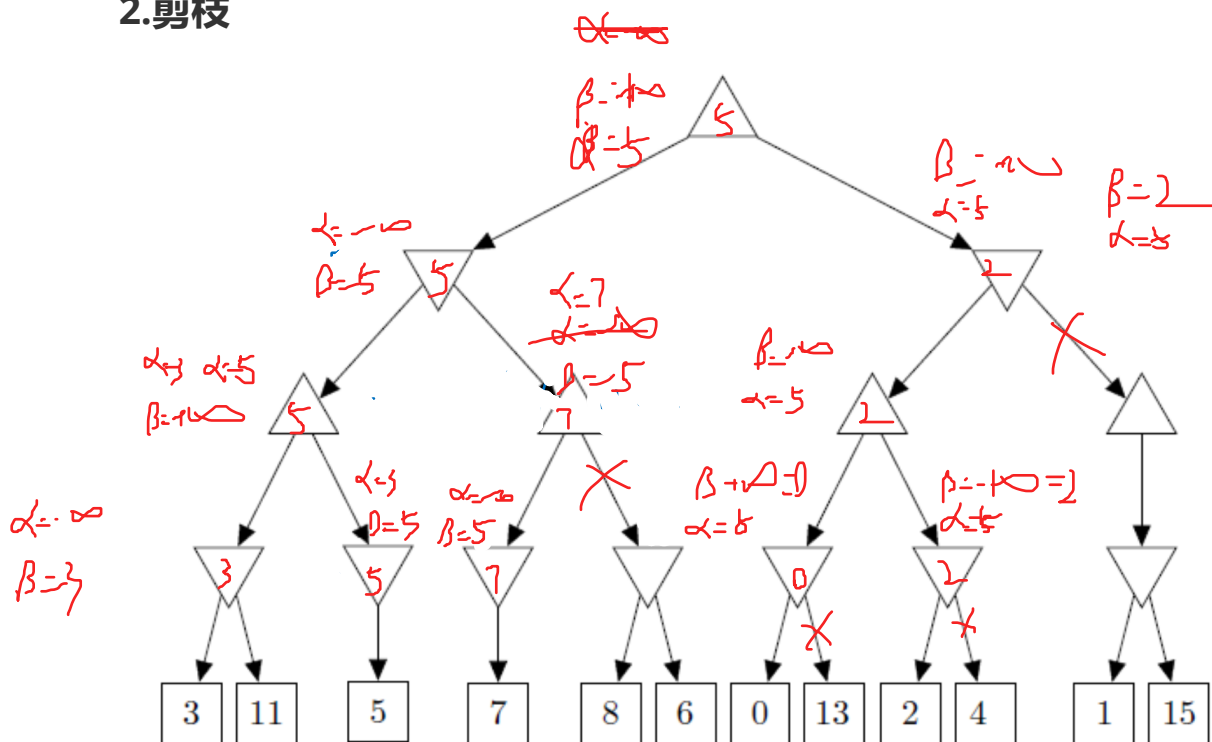


1.Search

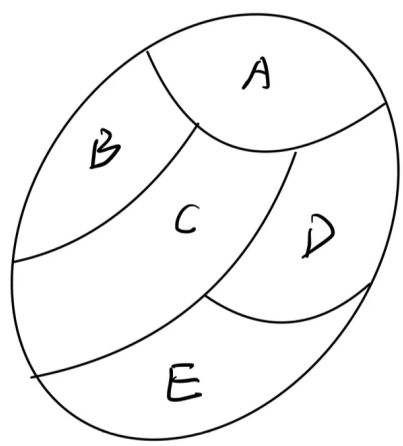

$$h(A) = 9 \quad h(B) = 3 \quad h(C) = 7 \quad h(D) = 4 \quad h(E) = 3 \quad h(F) = 0$$

- (1) Best first search
- (2) A* search
- (3) is h admissible

2.剪枝



3. CSP



着色问题 色彩空间: {R, B, Y}

- (1) draw constraint graph
- (2) A = R forward checking , the domain of B,C,D,E
- (3) A = R arc consistency , the domain of B,C,D,E

4. reasoning

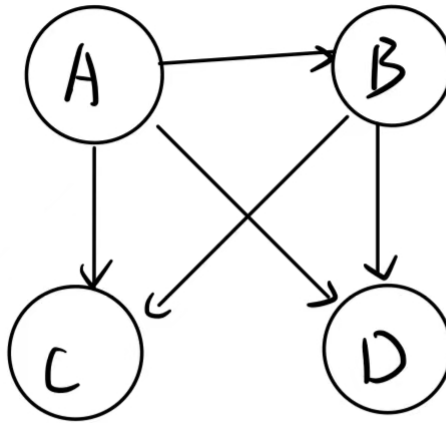
$R \Rightarrow U \quad U \Rightarrow (\neg W) \quad \neg R \Rightarrow \neg W$

Proof: $\neg W$

5. decision tree

A	B	C	Y(output)
F	F	F	F
T	F	T	T
T	T	F	T
T	T	T	F

6. Bayesian Network



概率（未特别说明指正确概率）

$$P(A) = 0.2$$

$P(B|A)$:

A	P(B)
T	0.8
F	0.7

$P(C|A,B) P(D|A,B)$

A	B	P(C)	P(D)
T	T	0.4	0.9
T	F	0.2	0.8
F	T	0.3	0.1
F	F	0.1	0.2

(1) 变量消元求 $P(c|\neg d)$

(2) C,D 条件独立性？ 给定A、 B

7. CNN

(1) input feature map $30 \times 30 \times 6$ 池化: 2×2 stride=2 问输出feature map尺寸

0 2	4 4
0 2	6 2
1 7	3 8
2 2	4 5

(2) 上述map, (1) 中池化层 maximum pool

(3) 同上average pool

8. MDP

A	B	C
D	E	F (food)

living reward =0 ;blocked by outer wall;

A = {North,East,South,West}

noise = 0

discount = 0.5

(1) optimal policy

state	$\pi(state)$
A	
B	
C	
D	
E	
F	

(2) Value iteration

k	A	B	C	D	E	F
0	0	0	0	0	0	0
1						
2						
3						
4						

(3) optimal value of A

which k iteration $V_k(A) = V^*(A)$